Questions

Q1.

A student rubs a plastic comb with a dry cloth to give the comb a positive electric charge.

Figure 19 shows the charged plastic comb picking up small pieces of paper.

(Source © GIPhotoStock/SCIENCE PHOTO LIBRARY)

Figure 19

(i) Explain how rubbing the comb with a dry cloth gives the comb a positive electric charge.

(3)

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(ii) Explain how the positively-charged plastic comb picks up the small pieces of paper.

(3)

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(Total for question = 6 marks)

Q2.

An atom contains electrons, neutrons and protons.
At a petrol station, a pipe is used to transfer petrol to the storage tanks.

The pipe is earthed.

There is friction between the petrol and the end of the pipe.

(i) Explain why it is dangerous **not** to earth the pipe.

(2)

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(ii) Explain how earthing the pipe makes this process much safer.

(2)

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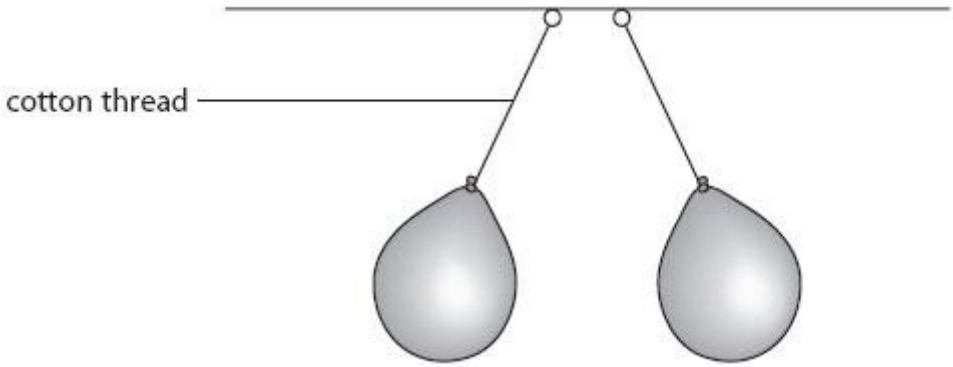
Q3.

(a) A student charges two balloons and hangs them side by side.



Explain why the cotton threads are not vertical.

(2)

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(b) The student rubs another balloon with a cloth.
This balloon becomes negatively charged.

(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

Compared to the charge gained by the balloon, the cloth gains

(1)

☐ **A** a larger negative charge

☐ **B** a larger positive charge

☐ **C** an equal negative charge

☐ **D** an equal positive charge

(ii) Explain why the balloon became negatively charged when it was rubbed with the cloth.

(2)

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(iii) The student then puts this charged balloon against a metal cabinet.

Describe what happens to the charge on the balloon where it touches the metal cabinet.

(2)

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(iv) The student charges another balloon and holds it against a wall. The charged balloon sticks to the wall when he lets go.

Suggest why the balloon is attracted to the wall.

(1)

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(Total for Question = 8 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about static electricity.

Figure 1 shows a plastic comb picking up small pieces of paper.

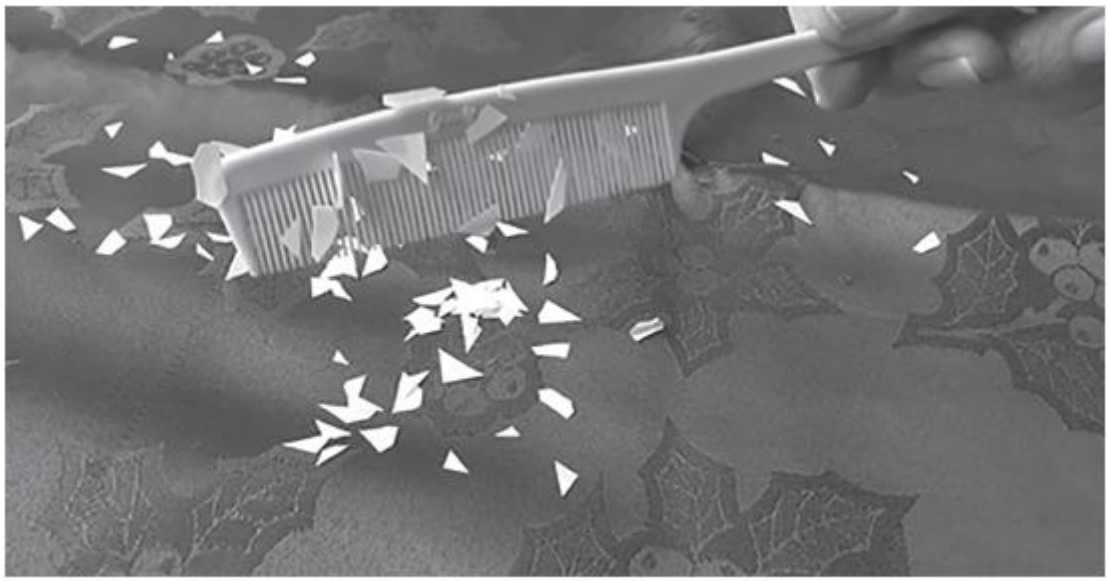

Figure 1

The comb picks up the paper because the comb has extra

(1)

- ☒ **A** magnetism
- ☒ **B** charge
- ☒ **C** resistance
- ☒ **D** weight

(Total for question = 1 mark)

Q5.

Figure 15 shows a metal chair being sprayed with paint.

The paint droplets come from a gun with an electric charge.

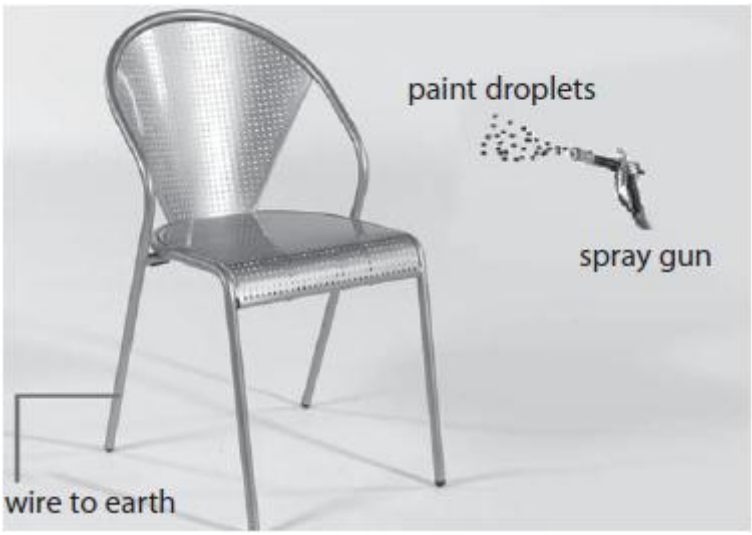

Figure 15

Inside the spray gun, electrons move along a charged wire towards the nozzle to charge the paint.

The charged paint droplets are sprayed from the nozzle.

The chair is connected to earth.

Which row of the table shows the correct combination of the charges as the charged paint droplets get near to the chair?

(1)

		paint droplets	chair
☐	**A**	negative	negative
☐	**B**	negative	positive
☐	**C**	positive	negative
☐	**D**	positive	positive

(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Figure 13 shows a negatively charged metal sphere, M.

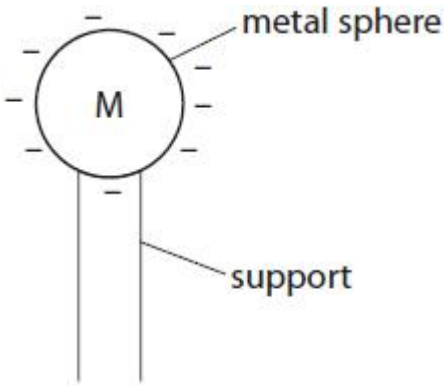

Figure 13

(i) Sphere M is negatively charged because it has

(1)

☐ **A** gained electrons
☐ **B** lost electrons
☐ **C** gained protons
☐ **D** lost protons

(ii) A metal sphere, N, is connected to earth by a wire.

N is moved near to M as shown in Figure 14.

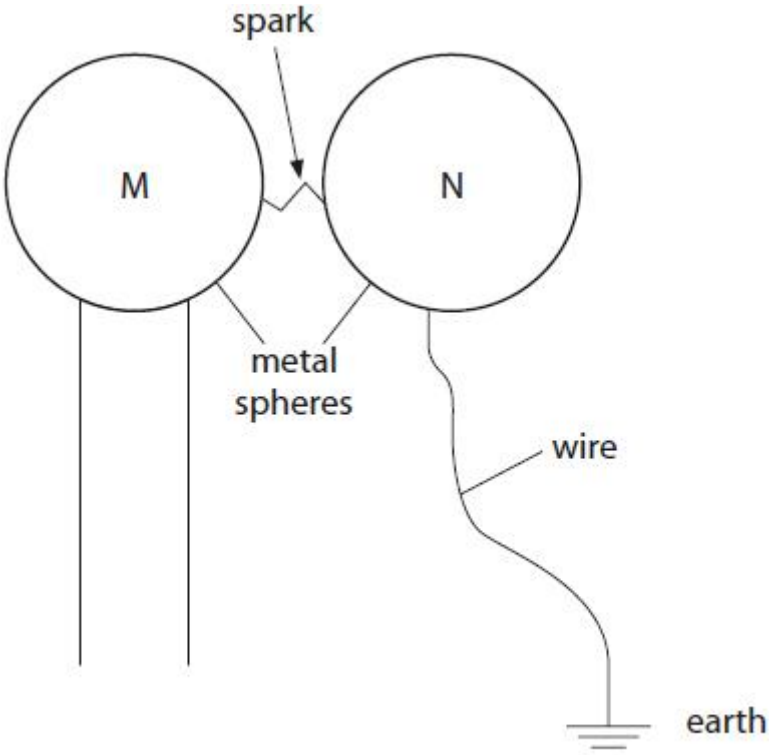

Figure 14

A spark jumps between the spheres, discharging sphere M.
Describe what happens in the wire connecting sphere N to earth when the spark jumps between M and N.

(2)

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(iii) Describe a use of earthing in everyday life.

Your answer should state the use and describe why earthing is needed.

(2)

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(Total for question = 5 marks)

Q7.

Figure 20 shows two metal spheres.

Metal sphere A is fixed to a table.
Metal sphere B can be moved.

Metal sphere B is placed at a short distance from metal sphere A.

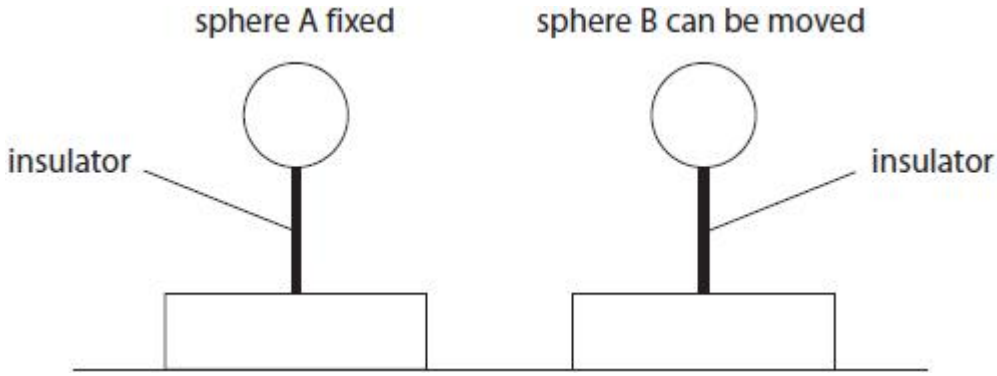

Figure 20

Both spheres are insulated from the table and given a negative charge.

The force between the charged spheres is measured.

(i) Explain, in terms of electric fields, why a force is exerted on sphere B.

(2)

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(ii) Sphere B is moved and the force between the spheres is measured at several different distances.
Figure 21 is a graph of force on sphere B against distance between the centres of the spheres.

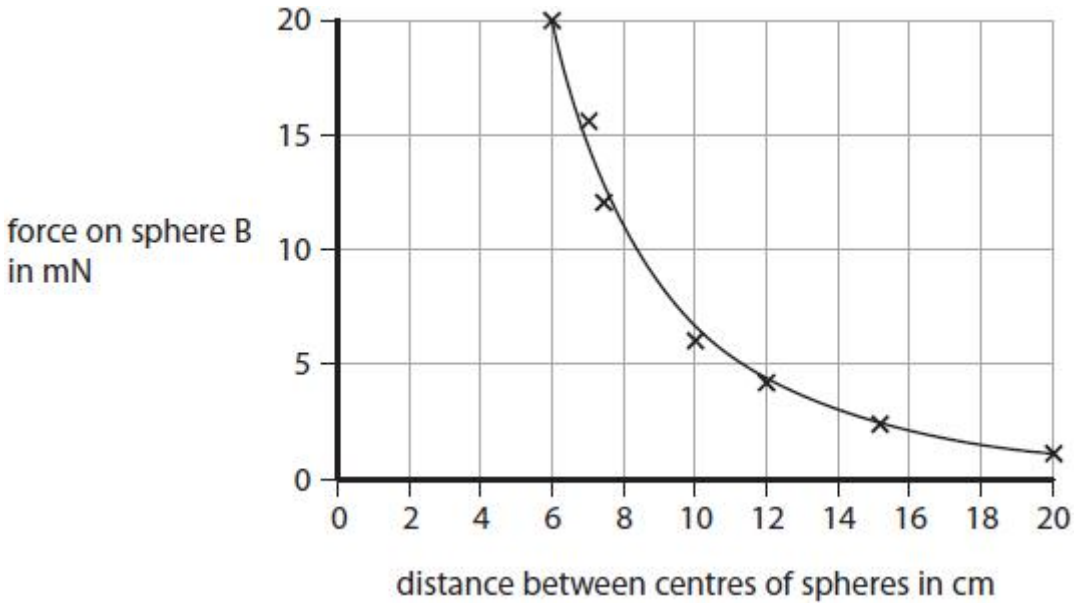

Figure 21

Describe how the force on sphere B varies with the distance between the centres of the spheres.

(2)

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(Total for question = 4 marks)

Q8.

* Figure 14 shows fuel being transferred to an aeroplane.

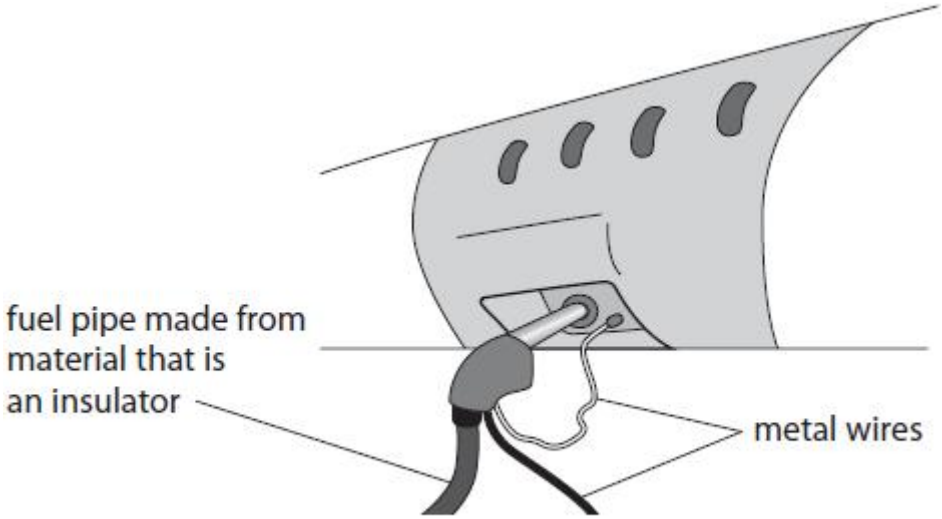

Figure 14

Explain why transferring fuel can be dangerous and how the use of metal wires makes the process much safer.

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(Total for question = 6 marks)

Q9.

Figure 18 shows an aeroplane being refuelled.

The fuel passes through a pipe into fuel tanks in the wings.

The fuel is very flammable.

(Source: © Stanisław Tokarski/123RF)

Figure 18

A build-up of static electric charge can be dangerous while the aeroplane is being refuelled.

Explain

- why the build-up of static electric charge would be dangerous
- how the danger is reduced.

(6)

(Total for question = 6 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about static electricity.

Figure 26 shows a plastic block and a metal disc with an insulating handle.

The top surface of the plastic block has a negative charge.

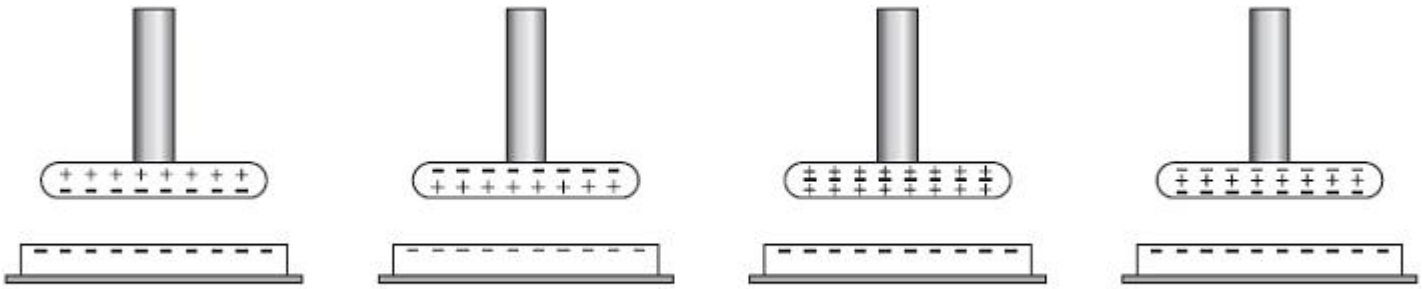
The metal disc has no overall electric charge.

Figure 26

A student uses the insulating handle to hold the metal disc above the plastic block.

(i) Which of these diagrams shows how the charge is distributed on the metal disc?

(1)



☐ **A** ☐ **B** ☐ **C** ☐ **D**

(ii) The student keeps holding the metal disc above the charged plastic block and taps the metal disc with a finger.

This earths the metal disc for a short time.
Explain why the disc now has an overall positive charge.

(2)

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(iii) Figure 27 shows the charges on part of the metal disc and the plastic block.

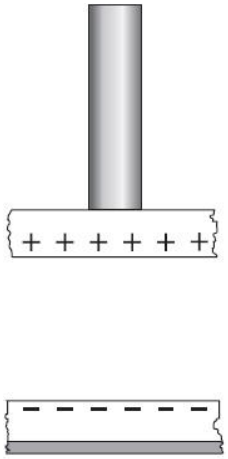

Figure 27

On Figure 27, draw lines to show the shape and direction of the electric field between the metal disc and the plastic block.

(2)

(Total for question = 5 marks)

Q11.

Glass is an insulator.

A student rubs a piece of glass with some silk.

The glass becomes positively charged.

(i) Explain how rubbing silk charges the glass.

(2)

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(ii) The silk is also charged when it rubs against the glass.
Describe the charge on the silk.

(2)

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(Total for question = 4 marks)

Mark Scheme

Q1.

Question Number:	Answer	Additional guidance	Mark
(i)	an explanation linking 3 of the following:		(3) AO 2 1
		reference to positive electrons or positive charge moving loses that mark point	
	friction (between cloth and comb) (1)		
	transfer of electrons / charge {from plastic comb / on to the cloth} (1)	electrons/charges are rubbed off comb (on to cloth)	
	electrons carry a negative charge (1)	leaving cloth with negative charge	
	leaving **excess** positive charge on the comb (1)	more protons than electrons (on the comb)	

Question Number:	Answer	Additional guidance	Mark
(ii)	an explanation linking:		(3) AO 2 1
	a negative charge is induced (1)	allow a clear description of induction	
		ignore references to positive charge being moved in this context only	
	on the part of the paper closest to the comb (1)		
	opposite charges attract (1)	force of attraction sufficient to pick up the pieces of paper	

Q2.

	Answer	Acceptable answers	Mark
(i)	An explanation linking • (friction/it) produces charges (at the end of the pipe) (1) • charge jumps to fuel tank (1) • (charge/friction) causes a spark (1) • can cause a fire /explosion (1)	static (electricity) builds up	**(2)**
(ii)	An explanation linking • (excess) charge / electrons (1) • Removed/ conducts away (1)	static charge discharged/ neutralised discharge current scores both marks	**(2)**

Q3.

	Answer	Acceptable answers	Mark
(a)	an explanation linking: balloons repel (1) (because) they have like charges (1)	balloons repulse / push away (from each other/to the side) same charge / both positive / both negative accept like charges repel for 2 marks	**(2)**
(b)(i)	☒ **D** an equal positive charge		**(1)**
(b)(ii)	an explanation linking any two of friction (between cloth and balloon) (1) transfer of electrons (1) (electrons/negative charges move) from cloth to balloon (1)	charge/electrons move accept balloon gains electrons from the cloth for 2 marks	**(2)**
(b)(iii)	a description including **two** from the following: • balloon becomes discharged (1) • metal /cabinet is a conductor (1) • electrons move through / on to metal / cabinet (1)	earthed / neutral (negative) charge for electrons accept electrons move to earth for 2 marks	**(2)**

(b)(iv)	(surface of) wall (becomes) positively charged /charged by induction (1)	charges on the wall separate charge closest to the surface of the wall is opposite to the charge on the balloon	**(1)**

Q4.

Question number	Answer	Additional guidance	Mark
	B charge A, C and D are incorrect force or property associations		**(1)** **AO1**

Q5.

Question number	Answer	Mark
	B negative positive	**(1)**

Q6.

Question number	Answer	Mark
i	A gained electrons Options B, C and D are incorrect explanations	**(1)** **AO1**

Question number	Answer	Additional guidance	Mark
ii	A description to include electrons / negative charges **move** (1) down the wire / to earth (1)		**(2)** **AO1**

<table>
<tr><th>Question number</th><th>Answer</th><th>Additional guidance</th><th>Mark</th></tr>
<tr><td>iii</td><td>A description including one row from:
<table>
<tr><th>use</th><th>description</th></tr>
<tr><td>fuelling cars / plane (1)</td><td>charge / voltage could build up causing a spark / fire: (avoided by earthing (the pump) (1)</td></tr>
<tr><td>(insecticide / paint) sprayers (1)</td><td>earthed object gains (induced) charge(s) to attract paint / insecticide (1)</td></tr>
<tr><td>kettle / other electrical device (1)</td><td>earthing the outside prevents shock (to user) (1)</td></tr>
</table></td><td>Other examples are possible

in this context also allow to prevent shock</td><td>(2)
AO1</td></tr>
</table>

Q7.

Question Number:	Answer	Additional guidance	Mark
(i)	An explanation linking: sphere A has an electric field (1) sphere B is in it (1)	 both spheres have electric fields the electric fields interact/overlap ignore nature of force; e.g. repulsion	(2) AO 2 2

Question Number:	Answer	Additional guidance	Mark
(ii)	a description to include: as the distance increases the force (on the sphere B) decreases (1) the greatest change is at smallest distances (1)	 negative correlation non-linear gradient changes allow named non-linear functions such as exponential / inversely proportional in this context reference to inverse square law scores 2 marks	**(2)** AO 3 1a AO 3 1b

Q8.

Question Number	Answer	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1 6 marks** **dangers**	**(6)**
	• friction as fuel flows through pipe • build-up of (electrostatic) charge • potential difference between nozzle and plane • causes spark • explosion or fire **use of metal wire** • potential is the same on both objects • no electric field • earths excess charge • constant safe discharge • no imbalance of electrons	

Descriptor
• No rewardable material.
• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels Eg - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Two unlinked statements	Possible candidate responses make a spark/ explosion/fire there is static electricity fuel is flammable metal wires conduct charge(electricity) could get an electric shock
Level 2	3–4	Additional guidance Limited explanation **linking** facts about dangers OR **linking** facts about why using metal wires is safer	Possible candidate responses A spark is produced because there is a build up of static charge (electricity) or build up of static charge prevented(electricity)because the metal wire takes the charge to earth(ground)
Level 3	5–6	Additional guidance Detailed explanation about dangers AND why using metal wires is safer (one may be stronger than the other but both should feature for level 3)	Possible candidate responses Spark is caused by the build up of charge (static electricity) AND the build up is prevented by the metal wire taking the charge to earth (ground)

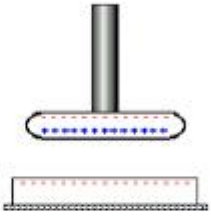
Q9.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. Description of danger • build-up of charge could create a spark • flammable fuel can easily catch fire • spark could ignite fuel • igniting fuel could cause a fire / explosion of the plane Description of how risk is reduced • The pipe and the airplane are connected by a metal wire • The metal wire is connected to ground/earth • Pipe and airplane at same potential • Metal is a conductor • Electrons can move through metal wire • No charge build-up • No danger of spark • Reduce charge separation by plausible method such as reduce flow rate/ wider pipe / less friction	**(6)** **AO1**

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) • Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) • Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) • Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Two statements	Possible candidate responses There could be a fire that could lead to an explosion.
Level 2	3–4	Additional guidance limited explanation **linking** facts about dangers arising from charge OR **linking** facts about how danger is reduced	Possible candidate responses A spark could cause a fire and explosion. OR Build-up of charge prevented by a wire connected to ground
Level 3	5–6	Additional guidance Detailed explanation about danger AND how danger is reduced. (one may be more detailed than the other but both should be present)	Possible candidate responses There could be a spark that could cause a fire in the fuel and explode. AND Wires between the airplane, pipe and ground prevent the build-up of charge.

Q10.

Question number	Answer	Additional guidance	Mark
(i)	B  A and D are incorrect because a negative charge cannot induce a negative charge C is incorrect because the disc is insulated so negative charge cannot be removed		(1)

Question number	Answer	Additional guidance	Mark
(ii)	an explanation linking: electrons / negative charges have flowed (1) from the metal disc / to the student / to earth/ground (1)	 reject positive charge moving for first mark	(2)

Question number	Answer	Additional guidance	Mark
(iii)	at least three straight lines joining disc and plastic (1) arrow(s) from disc towards plastic (1)	judge by eye ignore curved lines at edge do not award mark if there are arrows in both directions	(2)

Q11.

Question number	Answer	Additional guidance	Mark
(i)	An explanation that combines identification - understanding (1 mark) and reasoning - understanding (1 mark): charges move (1) because of friction (1)	 (negative) electrons transfer glass loses electrons	(2)

Question number	Answer	Mark
(ii)	An explanation that combines identification understanding (1 mark) and reasoning - understanding (1 mark): (negative) electrons are rubbed off the glass (on to the silk) (1) giving the silk a negative charge (1)	(2)